EXPERIMENT 1 (HARIN T_192412412)

1. Aim

To simulate and analyze TE_{10} and TM_{11} modes in a WR-90 rectangular waveguide using MATLAB and visualize their electric-field and magnetic-field distributions.

2. Apparatus Required

1. **Computer or laptop**
2. **MATLAB software**

3. Input Parameters

Parameter	Value
Operating frequency	10 GHz
Waveguide width	22.86 mm
Waveguide height	10.16 mm
Waveguide length	100 mm
Modes	TE_{10} and TM_{11}

4. Procedure

1. **Open MATLAB and create a new script.**
2. **Enter the operating frequency and waveguide dimensions.**
3. **Calculate the cutoff frequencies and propagation characteristics of the selected modes.**
4. **Generate a two-dimensional grid representing the waveguide cross-section.**
5. **Plot the electric-field and magnetic-field distributions for TE_{10} and TM_{11} modes.**
6. **Plot the variation of the TE_{10} propagation constant with frequency.**
7. **Observe the field patterns and compare the propagation behavior of both modes.**

5. MATLAB Program

```
clc; clear; close all;

c=3e8; f=10e9;

a=22.86e-3; b=10.16e-3;

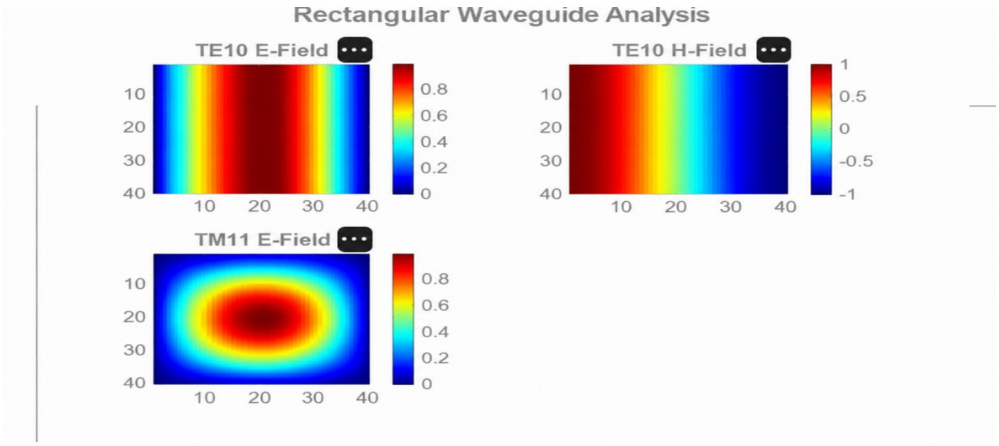
fc=c/(2*a);

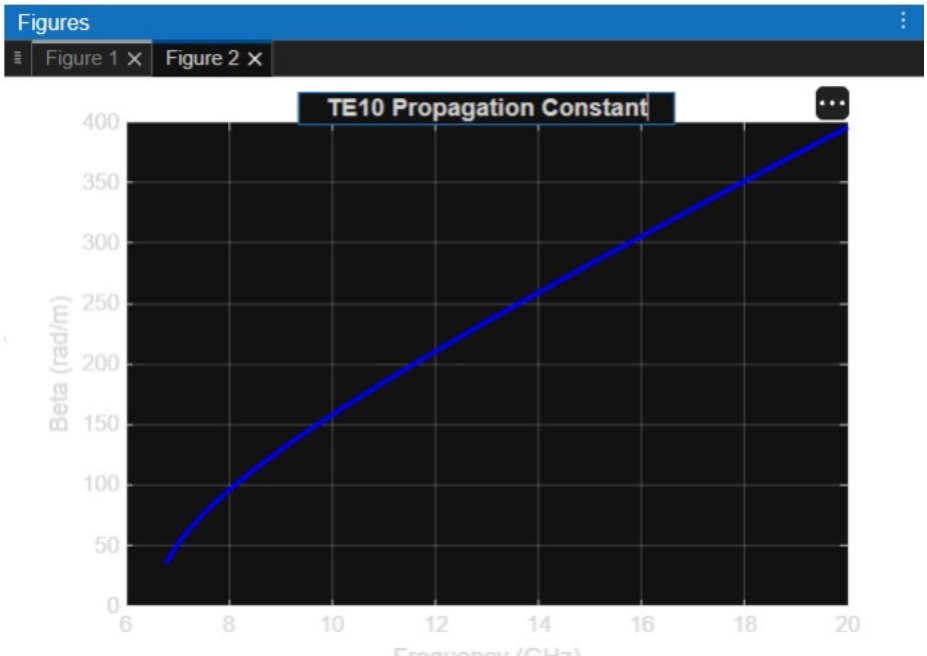
ft=c/2*sqrt((1/a)^2+(1/b)^2);
```

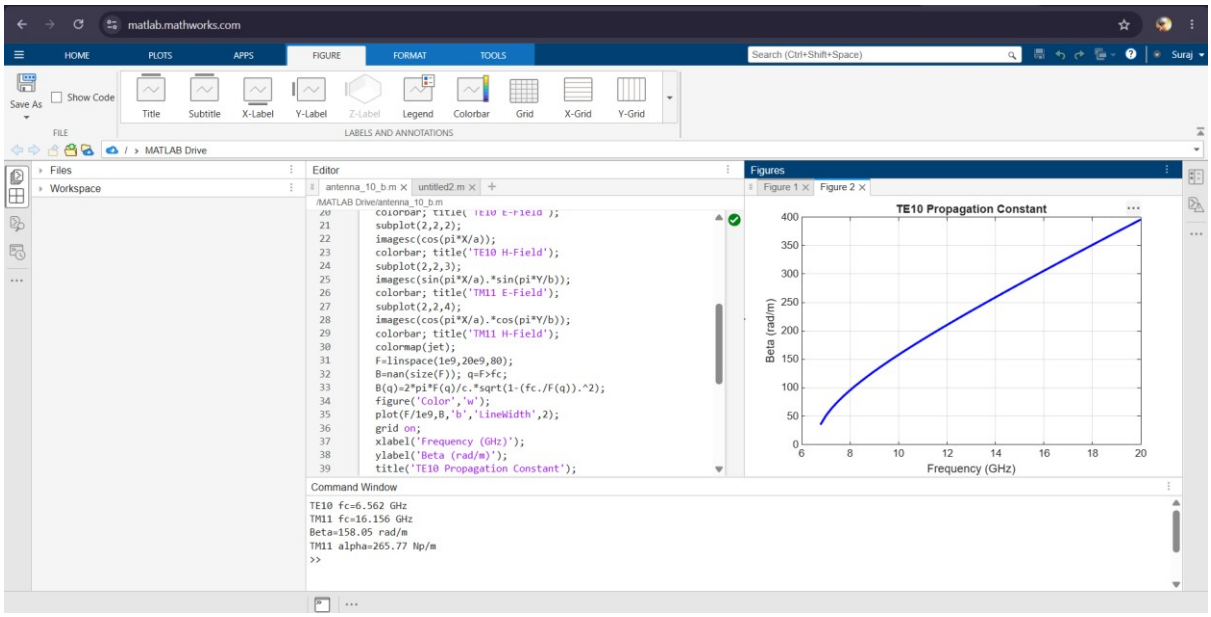
```
k=2*pi*f/c;
beta=sqrt(k^2-(2*pi*fc/c)^2);
al=sqrt((2*pi*ft/c)^2-k^2);
fprintf('TE10 fc=%.3f GHz\n',fc/1e9);
fprintf('TM11 fc=%.3f GHz\n',ft/1e9);
fprintf('Beta=%.2f rad/m\n',beta);
fprintf('TM11 alpha=%.2f Np/m\n',al);
x=linspace(0,a,40);
y=linspace(0,b,40);
[X,Y]=meshgrid(x,y);
figure('Color','w');
sgtitle('Rectangular Waveguide Analysis');
subplot(2,2,1);
imagesc(sin(pi*X/a));
colorbar; title('TE10 E-Field');
subplot(2,2,2);
imagesc(cos(pi*X/a));
colorbar; title('TE10 H-Field');
subplot(2,2,3);
imagesc(sin(pi*X/a).*sin(pi*Y/b));
colorbar; title('TM11 E-Field');
subplot(2,2,4);
imagesc(cos(pi*X/a).*cos(pi*Y/b));
colorbar; title('TM11 H-Field');
colormap(jet);
F=linspace(1e9,20e9,80);
B=nan(size(F)); q=F>fc;
B(q)=2*pi*F(q)/c.*sqrt(1-(fc./F(q)).^2);
figure('Color','w');
plot(F/1e9,B,'b','LineWidth',2);
```

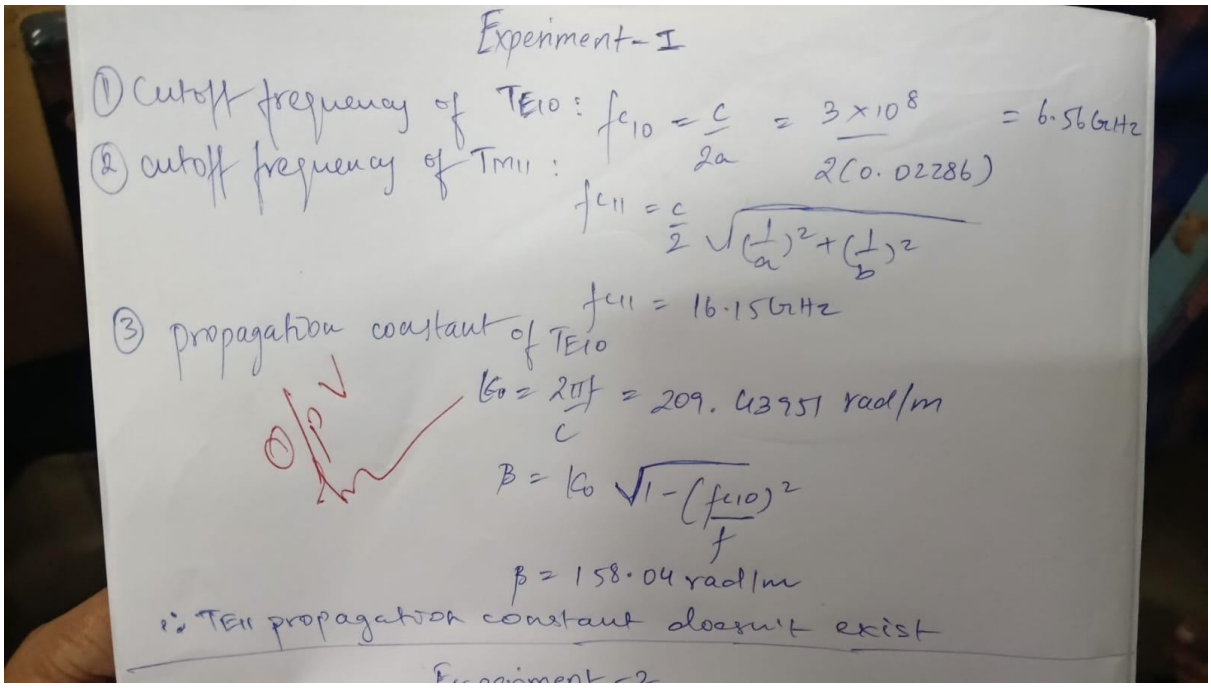
```
grid on;
xlabel('Frequency (GHz)');
ylabel('Beta (rad/m)');
title('TE10 Propagation Constant');
```

OUTPUT:









6. Reflection

I learned how electromagnetic waves propagate through rectangular waveguides in different modes. I understood how to visualize electric and magnetic field distributions using MATLAB and compare the propagation characteristics of TE_{10} and TM_{11} modes.

7. Result

The TE_{10} and TM_{11} modes in a WR-90 rectangular waveguide were analyzed using MATLAB. The electric-field and magnetic-field distributions were plotted, and the propagation characteristics were studied. The TE_{10} mode propagates at 10 GHz, whereas the TM_{11} mode is below cutoff and cannot propagate as a travelling wave.